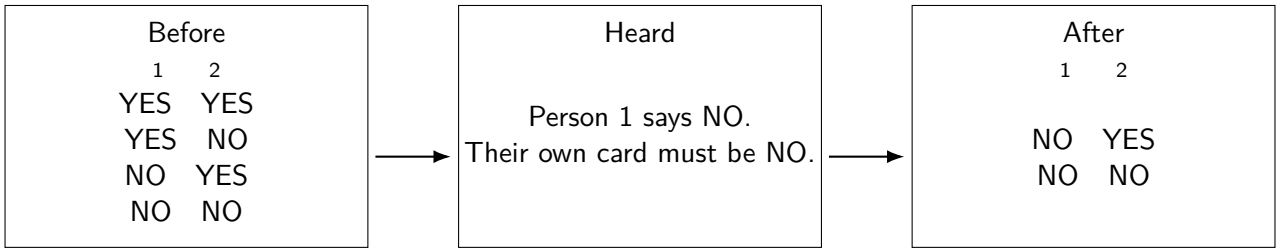
Each person sees only their own card: YES if they want a cup, NO if not. Everyone knows the cards can be dealt in any way.

In order, people 1, 2, 3 answer "Does everyone want a cup?" Say YES or NO when your card and the earlier answers settle it. Otherwise say NOT YET.

With two people:



Problem 1: Find every deal card that fits each conversation. Start with all eight deal cards each time.

Conversation 1

1 NOT YET

2 NOT YET

3 NO

Conversation 2

1 NO

2 NO

3 NO

Conversation 3

1 NOT YET

2 NO

3 NO

Conversation 4

1 NOT YET

2 NOT YET

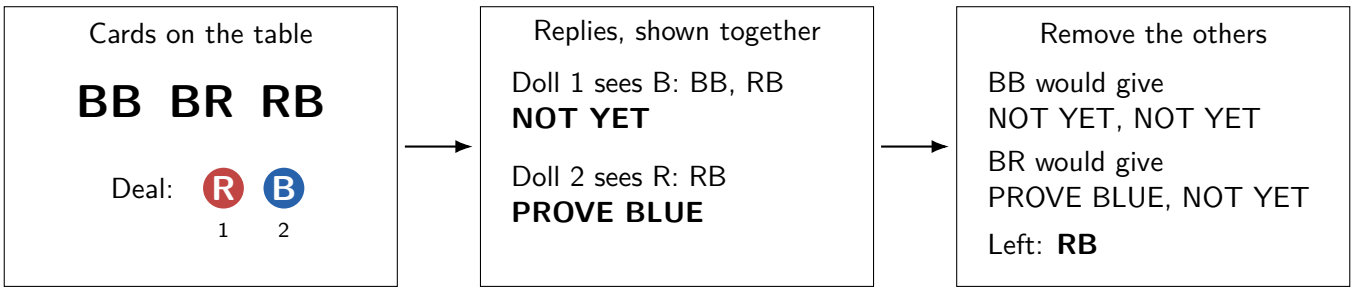
3 YES

Each doll wears B (blue) or R (red), sees the other dolls' headbands and cannot see its own. Everyone hears: "At least one headband is blue."

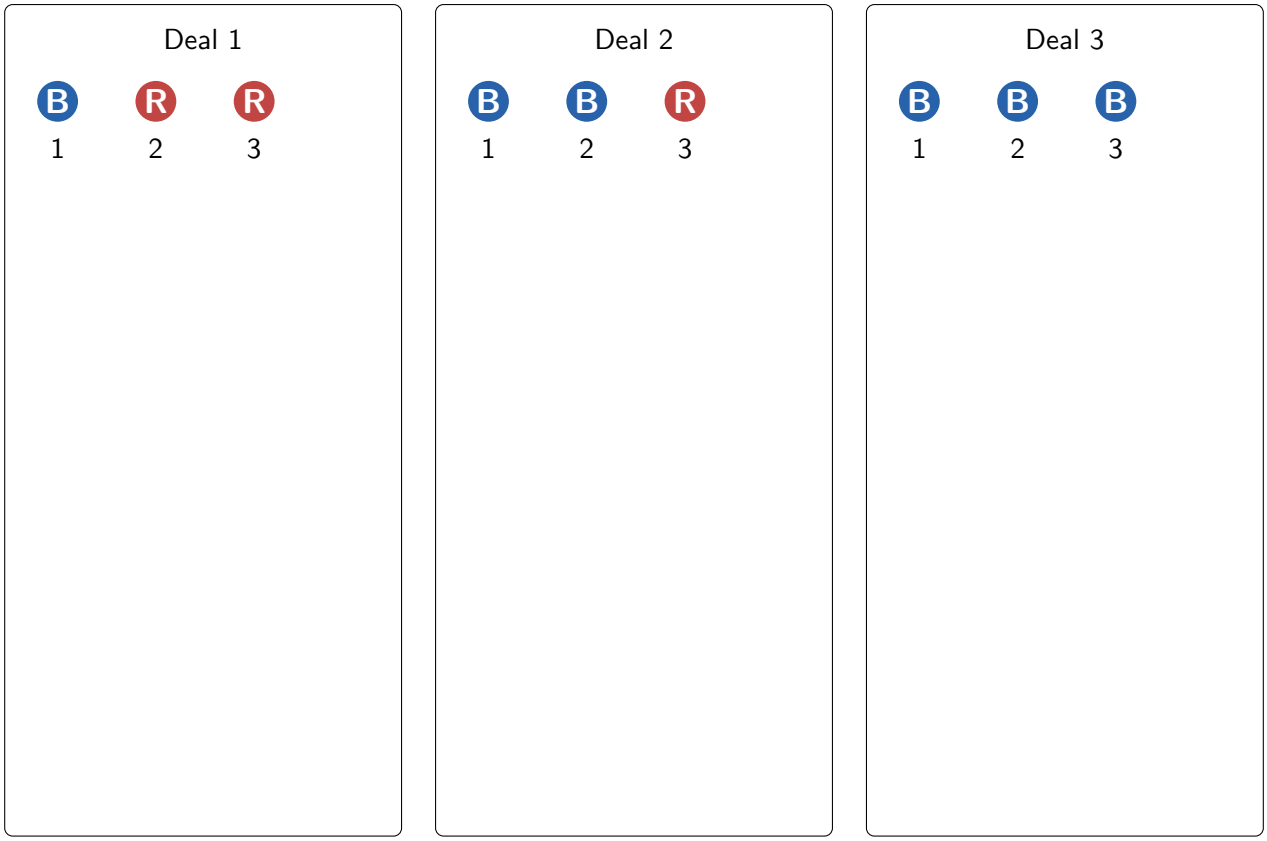
A doll shows PROVE BLUE only when every possible arrangement matching what it sees gives it a blue headband. Otherwise it shows NOT YET.

Choose all three replies from the cards on the table, without removing any. Show them together. Then remove every card that would have given different replies. Keep the headbands unchanged.

With two dolls:



Problem 2: Coach three dolls with each of these deals. Find their replies in each round until every blue doll can show PROVE BLUE. Start each deal with the eight headband cards, then use the shared announcement.



Problem 3: Start each deal with the announcement and all seven allowed headband cards. Put deals into groups in which each doll gives the same reply in round 1. Can later rounds separate every group?

Problem 4: Use all eight headband cards and make no announcement. Can any doll ever show PROVE BLUE from these rounds? Settle it with the cards.

1	2	3
YES	**YES**	**YES**

1	2	3
YES	**YES**	**NO**

1	2	3
YES	**NO**	**YES**

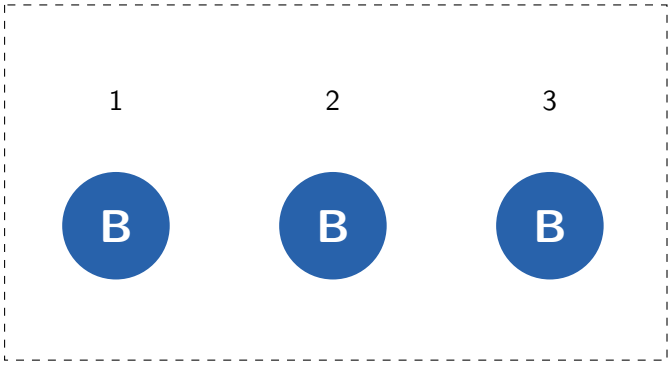
1	2	3
YES	**NO**	**NO**

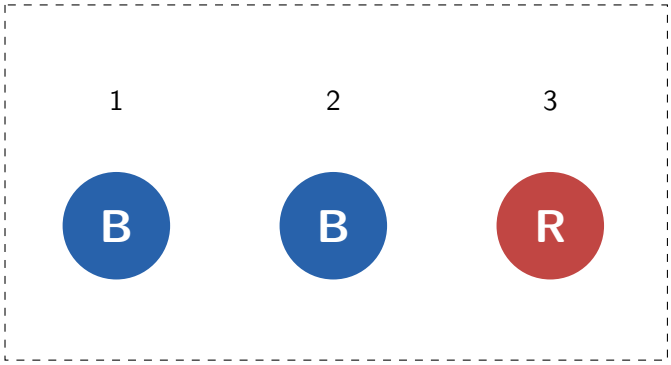
1	2	3
NO	**YES**	**YES**

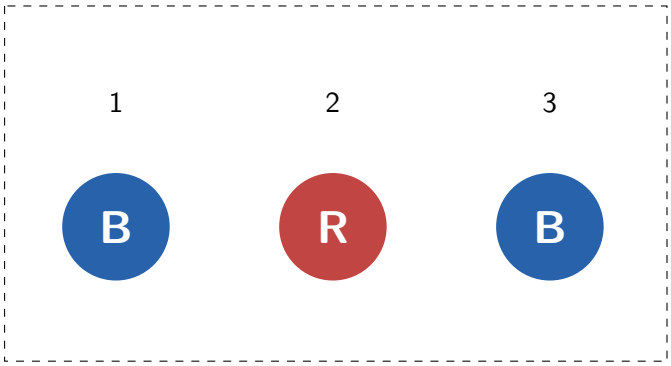
1	2	3
NO	**YES**	**NO**

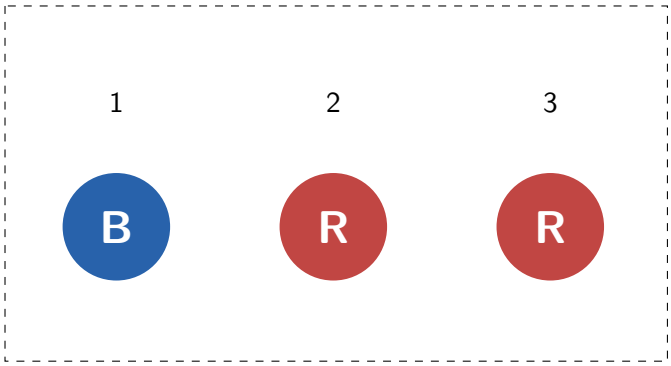
1	2	3
NO	**NO**	**YES**

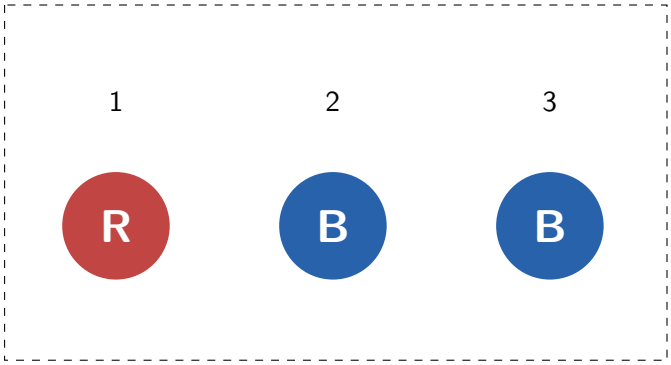
1	2	3
NO	**NO**	**NO**

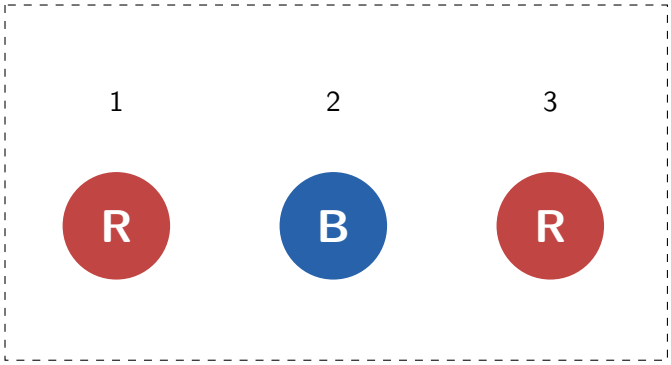

1
2
3
B
B
B

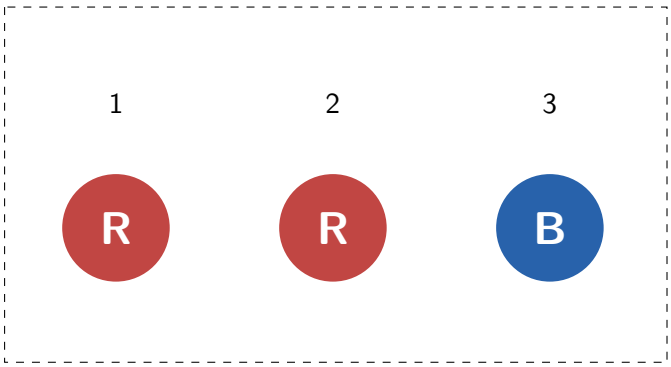

1
2
3
B
B
R

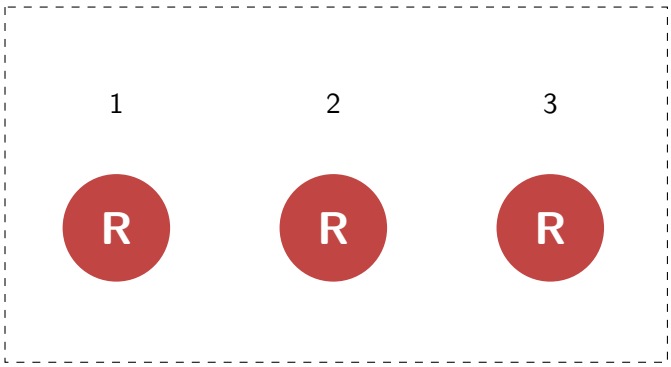

1
2
3
B
R
B

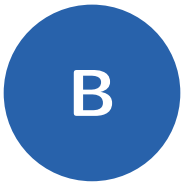

1
2
3
B
R
R


1
2
3
R
B
B


1
2
3
R
B
R


1
2
3
R
R
B


1
2
3
R
R
R

YES	YES	YES
NO	NO	NO
 BLUE	 BLUE	 BLUE
 RED	 RED	 RED

PROVE BLUE

PROVE BLUE

PROVE BLUE

NOT YET

NOT YET

NOT YET

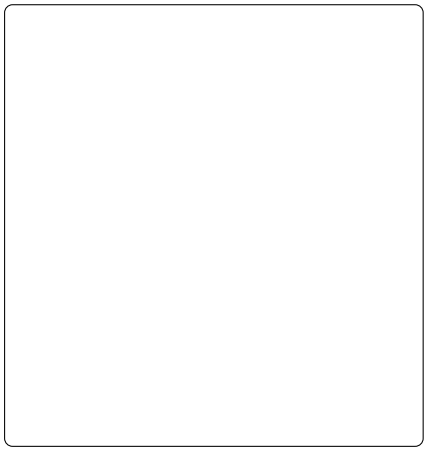
YES

NO

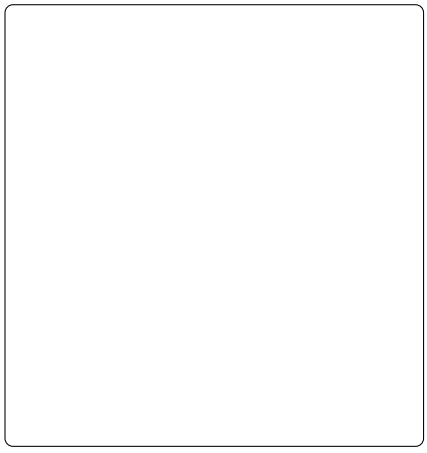
Shared possibilities

Publicly ruled out

Doll 1



Doll 2



Doll 3

